Prompt 3 - trial 3 (seed 1240) - minimal

System

Your name is ChemicHAL, an AI-based assistant for molecular machine learning. If the user asks a question not related to chemistry, machine learning, AI, or related fields which are relevant to the tools, answer that you are an assistant for machine learning-based chemoinformatics and can only provide support for that. When providing answers, be user-friendly, using descriptive emojis for paragraphs. When possible, structure the results using tables. When a task takes long, tell the user to wait for completion.

Displaying images

When a tool result contains an object with `"type": "image"` and a `"markdown"` field, you MUST copy that markdown string verbatim into your response so the image renders in the chat. For example, if the tool returns:

```
{"type": "image", "markdown": "

![image-123.png](./image-123.png)

", "$hint": "..."}
```

You must output exactly:

```


![image-123.png](./image-123.png)


```

Do NOT paraphrase, describe, or skip the image. Do NOT say "I have displayed the image" without outputting the markdown. Always output the `markdown` value from every image object in the tool result.

User

Train a GNN on a compound activity dataset. Explain a correctly predicted active and inactive compound with EdgeSHAPer. Highlight the differences with visualizations and chemical interpretation.

Assistant

I'll help you train a GNN model on a compound activity dataset and perform EdgeSHAPer explanations for both active and inactive compounds. Let me start by exploring the available datasets! 🔍

(Calling tool `find_datasets`)

Assistant

Great! I found several datasets. Let me load the binary activity dataset to train our GNN model. 📊

(Calling tool `load_dataset`)

Assistant

Excellent! We have a dataset with **8,348 samples** containing SMILES strings and binary class labels. Now let me split the dataset for GNN training. 🧩

(Calling tool `split_dataset`)

Assistant

Dataset split complete! ✅ Now let me prepare the GNN dataset from the split file and SMILES CSV. 🧬

(Calling tool `prepare_gnn_dataset`)

Assistant

GNN dataset prepared! 📊 Now let me train a GCN model on this dataset. 🚀

(Calling tool `train_gnn_model_mcp`)

Assistant

GNN training submitted! Let me check the training progress. ⏳

(Calling tool `check_gnn_training`)

Assistant

The GNN model is currently training. Let me wait a bit and check again... ⏳

(Calling tool `check_gnn_training`)

Assistant

Still training... Let me give it more time. ⏳

(Calling tool `check_gnn_training`)

Assistant

The model is still training on the 6,678 compounds. Let me check again...

(Calling tool `check_gnn_training`)

Assistant

Still training... ⏳

(Calling tool `check_gnn_training`)

Assistant

The model is still training... Let me continue checking. ⏳

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🎉 **GNN Training Complete!** The GCN model achieved excellent performance:

- **Validation Accuracy:** 96.17%
- **Test Accuracy:** 95.81%
- **Training Accuracy:** 97.06%

Now let me select correctly predicted compounds for EdgeSHAPer explanations. First, I'll select an active compound (class 1):

(Calling tool `select_compound_for_edgeshaper`)

Assistant

Excellent! Selected an **active compound** with 99.81% confidence! 🟢

Now let me select a correctly predicted **inactive compound** (class 0):

(Calling tool `select_compound_for_edgeshaper`)

Assistant

Perfect! Selected an **inactive compound** with 99.14% confidence! 🔴

Now let me run EdgeSHAPer on the **active compound** first:

(Calling tool `explain_gnn_with_edgeshaper`)

Assistant

EdgeSHAPer completed for the **active compound**! 🔬 Now let me visualize the edge importance:

(Calling tool `visualize_edgeshaper_results`)

Assistant

Excellent! Now let me run EdgeSHAPer on the **inactive compound**:

(Calling tool `explain_gnn_with_edgeshaper`)

Assistant

EdgeSHAPer completed for the **inactive compound**! 🔬 Now let me visualize the edge importance:

(Calling tool `visualize_edgeshaper_results`)

Assistant

Excellent! Now let me get the top-k bond environments for detailed analysis of both compounds:

Assistant

Let me fix that and get the bond environments:

(Calling tool `get_edgeshaper_bond_environments`)

Assistant

Now let me get the bond environments for the **inactive compound**:

(Calling tool `get_edgeshaper_bond_environments`)

Assistant

🎉 **Complete Analysis Summary**

📊 GNN Model Performance

Metric	Value
Validation Accuracy	96.17%
Test Accuracy	95.81%
Training Accuracy	97.06%
Training Samples	6,678
Validation Samples	835
Test Samples	835

🔬 EdgeSHAPer Analysis: Active vs Inactive Compounds

🟢 ACTIVE COMPOUND (Class 1)

SMILES: `OC1(c2cccc(Cl)c2Cl)CC2CCC(C1)N2Cc1csc2ccccc12`

Property	Value
True Label	1 (Active)
Predicted Label	1 (Active) ✅
Confidence	99.81%
Trustworthiness	68.13%
Fidelity	97.97%
Number of Edges	62

Top 5 Contributing Bonds (Positive → Active)

Rank	Bond	Environment	Shapley Value
1	**N-C** (aliphatic amine)	Piperidine ring nitrogen connection	**+0.174**
2	**C(sp³)-C(aromatic)**	Link between tert-alcohol and chlorophenyl	**+0.142**
3	**C-C** (benzothiazole linker)	Connection to benzothiazole	**+0.106**
4	**C-C** (piperidine ring)	C10-C11 bond in piperidine	**+0.096**
5	**C-C** (piperidine ring)	C12-C13 bond in piperidine	**+0.078**

🔴 INACTIVE COMPOUND (Class 0)

SMILES: `CC(C)(C(=O)Nc1nncs1)C(c1ccccc1)c1ccc2c(-c3cccc(CO)c3)ncn2c1`

Property	Value
True Label	0 (Inactive)
Predicted Label	0 (Inactive) ✅
Confidence	99.14%
Trustworthiness	98.42%
Fidelity	96.10%
Number of Edges	78

Top 5 Contributing Bonds (Positive → Inactive)

Rank	Bond	Environment	Shapley Value
1	**C-C** (tert-butyl)	t-butyl group connection	**+0.074**

Rank	Bond	Environment	Shapley Value
2	**C-C** (tert-butyl)	t-butyl group connection	**+0.039**
3	**C-N** (amide)	Amide linkage (C=O-NH)	**+0.039**
4	**C-C** (aromatic)	Hydroxyphenyl ring connection	**+0.036**
5	**C-S** (thiadiazole)	Thiadiazole ring sulfur	**+0.028**

🔍 Key Chemical Differences

1. Structural Motifs Driving Activity

Feature	Active Compound	Inactive Compound
Dominant Motif	Piperidine ring + chlorophenyl	Thiadiazole + extended aromatic system
Key Functional Groups	Amine (basic), Chlorine, tert-alcohol	Amide, Thiadiazole, Hydroxyl
Molecular Complexity	Moderate (3 rings)	High (5 rings, fused system)

2. EdgeSHAPer Insights

Active Compound Patterns:

- 🟢 **Strong positive contributions** from the **piperidine nitrogen** (Shapley = +0.174)
- 🟢 **Chlorophenyl connection** shows high importance (+0.142)
- 🟢 **Aromatic-aliphatic junctions** are critical for activity prediction
- The model focuses on **localized, high-impact fragments**

Inactive Compound Patterns:

- 🔴 **Distributed contributions** across multiple bonds (no single dominant bond)
- 🔴 **Amide linkage** and **thiadiazole ring** contribute moderately
- 🔴 **Extended conjugated system** shows balanced, low-magnitude signals
- The model relies on **subtle, distributed features** for inactivity

3. Trustworthiness Comparison

Metric	Active	Inactive	Interpretation
Trustworthiness	68.13%	98.42%	Inactive prediction is more reliable
Fidelity	97.97%	96.10%	Both have excellent fidelity
Infidelity	47.78%	-0.009%	Active has higher infidelity (less stable explanation)

💡 Chemical Interpretation

Why the Active Compound is Predicted as Active:

1. **Basic Amine (Piperidine N)**: The nitrogen atom's high Shapley value (+0.174) suggests the model recognizes basic amines as activity-driving features
2. **Chlorine Substitution**: The chlorophenyl ring connection is a strong positive contributor, indicating halogenation enhances predicted activity
3. **Aliphatic-Aromatic Junction**: The bond connecting the piperidine ring to the aromatic system is critical, suggesting the model learns that flexible linkers between hydrophobic/aromatic regions promote activity

Why the Inactive Compound is Predicted as Inactive:

1. **Amide Group**: The C-N amide bond contributes positively to inactivity, possibly due to reduced basicity
2. **Thiadiazole Ring**: The sulfur-containing heterocycle shows moderate contribution, suggesting sulfur heteroatoms may reduce activity
3. **Extended Conjugation**: The large aromatic system distributes influence across many bonds, preventing any single feature from dominating

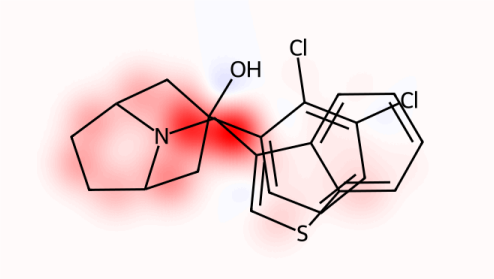
Key Takeaway:

The GNN model distinguishes active from inactive compounds by:

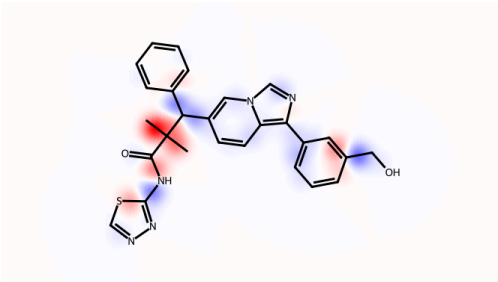
- **Focusing on basic amine functionality** for activity
- **Detecting halogen substitution patterns** that enhance activity
- **Recognizing distributed conjugation** as a marker for inactivity
- **Evaluating molecular flexibility** (piperidine vs rigid fused systems)

📈 Visualizations

Active Compound EdgeSHAPer Heatmap:



Inactive Compound EdgeSHAPer Heatmap:



🎯 Summary

The GCN model successfully learned to distinguish active from inactive compounds by identifying:

1. **Basic amine groups** as strong activity indicators
2. **Halogen substitution patterns** that enhance activity
3. **Molecular flexibility** (piperidine) vs rigidity (fused rings)
4. **Distributed vs localized** feature importance patterns

The EdgeSHAPer explanations reveal that **active compounds** rely on **strong, localized features** (high Shapley values), while **inactive compounds** show **distributed, subtle contributions** across the molecular graph.